

Customer Data Synchronization Platform

Lesson 5 Assessment — n8n Workflow Documentation

1. Assessment Objective

Build a production-style n8n customer data synchronization workflow that ingests customer records from API, database, and spreadsheet-style sources, normalizes email keys, merges records, aligns the final schema, handles missing values, validates profile completeness, and routes incomplete records to quarantine.

2. Workflow Architecture

Stage	Node / Branch	Purpose
Trigger	Start Sync	Starts the synchronization workflow.
Source 1	API Session Data → Normalize_Email	Provides session/device/activity data and normalizes email.
Source 2	DB Customer Profiles → Normalize_Email	Provides customer profile data and normalizes email.
Source 3	Sheet Marketing Prefs → Normalize_Email	Provides marketing preferences and normalizes email.
Merge 1	MERGE_API+Database	Combines API session data with database profiles using email.
Merge 2	MERGE_MarketingSheet	Combines the merged profile with marketing preferences using email.
Clean	CLEAN_CoalesceFields	Creates a consistent customer profile and applies fallback values.
Validation	IF_HasFullName	Checks whether the merged profile has the required full name.
Valid Output	OUTPUT_UnifiedCustomerProfiles	Sends complete profiles to the unified output.
Exception	Quarantine_UnmatchedProfiles	Routes incomplete profiles for review.

3. Multi-Source Ingestion

Three independent source branches are used: API session data, database customer profiles, and spreadsheet marketing preferences. Each branch contains mock customer records and is connected to the synchronization pipeline.

4. Email-Key Normalization

Email is the common matching key. Each source normalizes email values before merging by trimming leading/trailing whitespace and converting the value to lowercase. This reduces mismatches caused by different casing or accidental whitespace.

Normalization: trim → lowercase

5. Merge Strategy

The first merge combines API session data and database customer profiles using the normalized email field. The second merge combines that result with spreadsheet marketing preferences using the same normalized email key. This creates one synchronized customer record where source data can be combined by the shared identifier.

6. Schema Alignment and Coalescing

Unified Field	Design
email	Normalized email matching key.
fullName	Generated from firstName and lastName.
deviceType	Uses source value with a fallback when missing.
lastActive	Uses activity value with a fallback when missing.
marketingOptIn	Uses marketing preference with a safe default when missing.
preferredLanguage	Uses source language with a fallback when missing.

7. Validation and Quarantine

IF_HasFullName is the post-merge quality gate. Complete profiles continue to OUTPUT_UnifiedCustomerProfiles. Records that fail the completeness check are routed to Quarantine_UnmatchedProfiles so incomplete data does not enter the production-ready unified output.

8. Test Plan

Test	Expected Result
All three branches execute	API, database, and spreadsheet records reach the merge stages.
Different email casing	Values match after lowercase normalization.
Email with whitespace	Values match after trimming.
Missing optional fields	Fallback/default values are applied.
Complete customer profile	Record reaches unified customer output.
Missing fullName	Record is routed to quarantine.

9. Risk Assessment

Primary risks include inconsistent email formatting, missing source values, schema differences between systems, and incomplete customer profiles. The workflow mitigates these risks through email normalization, field coalescing/defaults, standardized output fields, and a post-merge quarantine path.

10. Setup / Reproduction Instructions

- Import the exported n8n workflow JSON into an n8n workspace.
- Review the three mock source branches and their sample records.
- Verify email normalization in all source branches.
- Verify both merge stages use email as the matching field.
- Run the workflow using Start Sync / Manual Trigger.
- Inspect unified profiles and quarantine output.
- Capture execution screenshots and export the final workflow JSON.

11. Assumptions and Limitations

The assessment uses mock customer datasets and native n8n workflow components. Source systems are represented by workflow data branches rather than production database/API credentials. The quarantine path is intended for review of incomplete profiles and can be extended with a ticketing or data-quality system in a production deployment.

12. Submission Evidence

Recommended evidence includes the complete workflow canvas, normalized email output, both merge outputs, clean/coalesced records, unified customer output, quarantine output, and a short demonstration showing the end-to-end execution.